

Privileged Session Management – Part 1

In this section, we will perform several tests to see the various privileged session management options that are available with **CyberArk PAM**.

First, we will disable the **PSM** globally and then activate it for specific platforms using exceptions. We will then perform tests to ensure that privileged session management is functioning properly using the various connection methods available:

- Privileged Session Manager (**PSM**) through the **PVWA**
- PSM for Windows
- PSM for SSH

Classic PSM

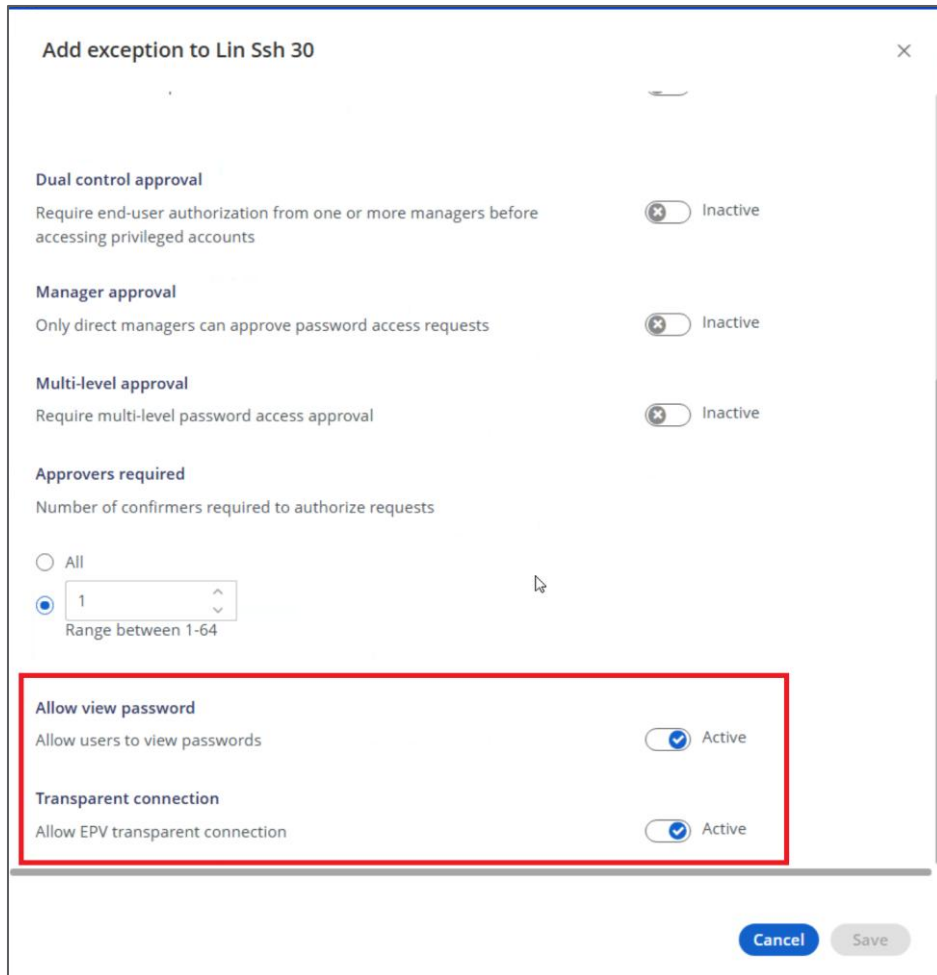
By **classic PSM**, we mean connecting through the **PVWA** to the **PSM** running on a Windows server, which is what we have been doing so far, activating the **PSM** globally for all active Platforms. In this section, we will see how to activate the **PSM** using exceptions, which will give us greater control over to what and how our users connect.

Remove Privileged Access Workflows Exceptions

To simplify the **PSM** testing, we will first disable the **Privileged Access Workflows** that we modified in earlier exercises. That way, we will not have dual control or exclusive access interrupting our testing. We can perform this action using the classic interface. Just bear in mind that this interface will probably be removed in a not-too-distant future release.

Note: Do **NOT** disable the Privileged Access Workflow **Allow EPV transparent connections**.

1. Log in to the **PVWA** as *Mike* using LDAP authentication.
2. Go to **Administration | Platform Management** and go to the platform **Lin Ssh 30**. Click on the ellipsis and select **Add exception**.
3. Go to the **Access workflow** tab and deactivate all the options **except** the last two:
 - **Allow users to view passwords**
 - **Allow EPV transparent connection**



4. Remember to click the **Save** button.
5. Do the same thing for the Platform **Svr Adm 20**.

Disabling the PSM Globally

As we saw in one of our first exercises, the **PSM** is enabled through the **Master Policy**. The **PSM** can be enabled either globally for all platforms or disabled globally and only activated through exceptions, which is what we will test here.

The **Master Policy** can now be edited in the new interface, so we will see how to do that in this exercise.

1. Login to the **PVWA** as *Mike* using LDAP authentication.
2. Go to **Administration | Platform management**. Drop down **Master policy** and select **Edit master policy**.



- Click on the **Session Management** tab and then switch the toggle for **Session monitoring** to **Inactive**. When you have finished, click **Save**.

- Remain in the **Platform Management** section.

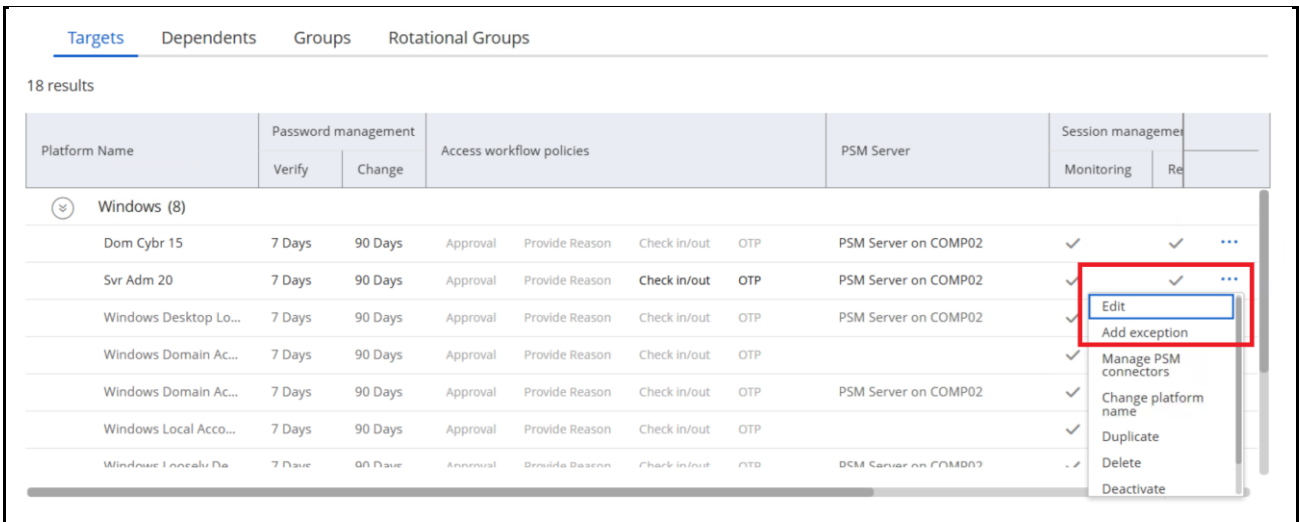
Activate the PSM by exceptions

This method enables users to connect via the **PSM** only to those accounts that are associated with Platforms that have been explicitly enabled by a Vault administrator. For instance, users will be able to connect to a target system using a *root* account that requires a logon account, but not directly via those logon accounts.

Adding Exceptions

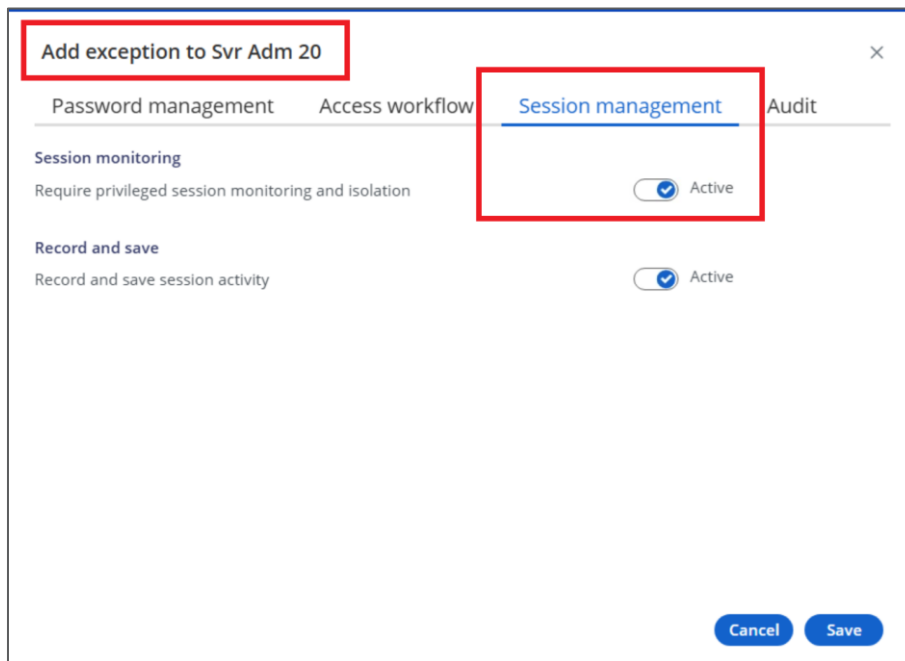
Let's activate the **PSM** by exception for our Platforms. We will begin with the Platform for local admins on Windows servers.

1. Go to **Administration | Platform Management**. Expand the **Windows** section under **Targets**. Then click on the ellipsis at the end of the line for **Svr Adm 20** and select **Add exception**.



The screenshot shows the 'Targets' tab in the Platform Management interface. A table lists 18 results, with a sub-section for 'Windows (8)'. The 'Svr Adm 20' target is highlighted. A context menu is open for this target, showing options like 'Edit', 'Add exception', 'Manage PSM connectors', 'Change platform name', 'Duplicate', 'Delete', and 'Deactivate'. The 'Add exception' option is highlighted with a red box.

2. Click on the **Session management** tab and then activate the toggle for **Session monitoring**.



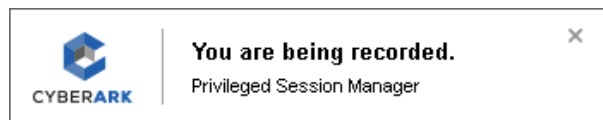
The screenshot shows a dialog box titled 'Add exception to Svr Adm 20'. It has four tabs: 'Password management', 'Access workflow', 'Session management', and 'Audit'. The 'Session management' tab is selected. Under 'Session monitoring', there is a toggle switch labeled 'Active' which is turned on. Below this, under 'Record and save', there is another toggle switch labeled 'Active' which is also turned on. At the bottom right, there are 'Cancel' and 'Save' buttons.

Repeat the above steps to enable the **PSM** for the **Lin Ssh 30** and **Ora Db a 30** platforms.

Connect with a Linux Account

In this first test, we will connect via the **PSM** exactly as we did previously, the only difference being that now the **PSM** is disabled by default and enabled by exception.

1. Login to the **PVWA** as *Carlos*, go to the **ACCOUNTS** page, and locate *root01*. Press the **Connect** button. As before, you will receive an RDP file. Click on it to launch the RDP session.
2. If everything was configured correctly, you should see a message that your session is being recorded.



3. And you will be connected to the target server.



4. Optionally, run some Linux commands.

```
mkdir test
rm -R test
```

5. Type **exit** to end the session.

Note: Notice that the connect button for the logon accounts are now greyed out.

Connect with an Oracle Account

1. Log out of the **PVWA** and log back in as the user *Robert*.
2. In the **Account view**, find the account *dba01* and click the **Connect** button.
3. Once again, you will see a message stating that your session is being recorded.

Note: If you receive a Remote Desktop Connect pop-up, “Your Remote Desktop Services session has ended”, retry the connection component. You may have to connect a couple of times before seeing the message.

4. In a later exercise, we will be searching for activity in this session by the commands that were run. Execute the following commands so that we will have some content for later:

```
select * from employees;
```

You will see a list of employees with their IDs, their salaries, and home cities.

Now let's give one of them a raise.

```
UPDATE employees set employee_salary = '75000' where employee_id = 0003;  
commit;  
exit
```

5. Type **exit** to end the session.

HTML5 Gateway

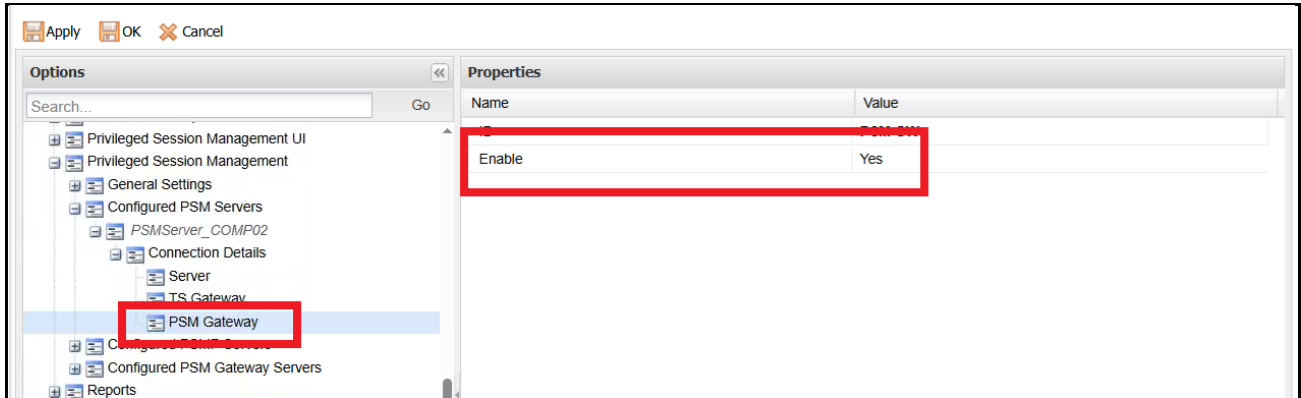
In this section, we will see how to configure the **PSM HTML5 Gateway**, which enables us to tunnel sessions between end users and the **PSM** server using a secure WebSocket protocol (port 443). This eliminates the need to open an RDP connection from the end user's machine. The RDP session is delivered to the end user through a browser tab, rather than via an RDP window.

Enable the HTML5 Gateway

Note: In this environment, the HTML5 Gateway has already been installed for you. It is running on **lin-451**, but it has not been enabled in the **PVWA**.



1. First, login to the **PVWA** as *Mike*, go to **ADMINISTRATION | Configuration Options | Options**.
2. Next, go to **Privileged Session Management | Configured PSM Servers | PSM_Server_COMP02 | Connection Details | PSM Gateway**.
3. Set the *Enable* parameter to **Yes** and click the **Apply** button.

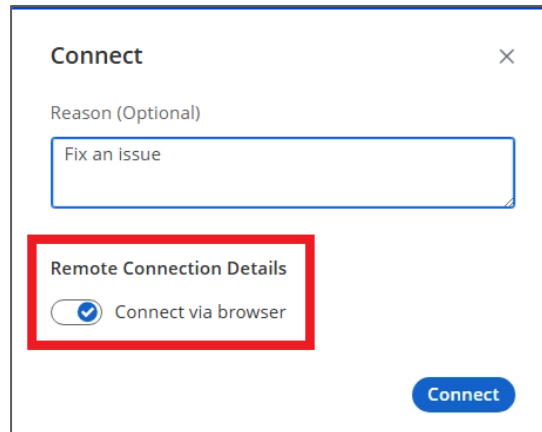


4. Click on **Apply** and then **OK** to save your changes.

Connect via HTML5 Gateway

In this section we will see how to connect to a target system through an RDP session delivered through your browser. We will also see how to transfer files from your local desktop to the remote system and the reverse.

1. Now log in as the user *John* and go back to the **ACCOUNTS** page and locate *administrator*.
2. Press the **Connect** button. This time, instead of downloading an RDP file, you will receive a pop-up asking whether you want to map your local drives (yes) and whether you want to *Connect using HTML5 GW* (also yes). Enable **Connect with your browser**. Optionally provide a reason for launching the connection and then click **Connect** to launch an HTML5 connection.



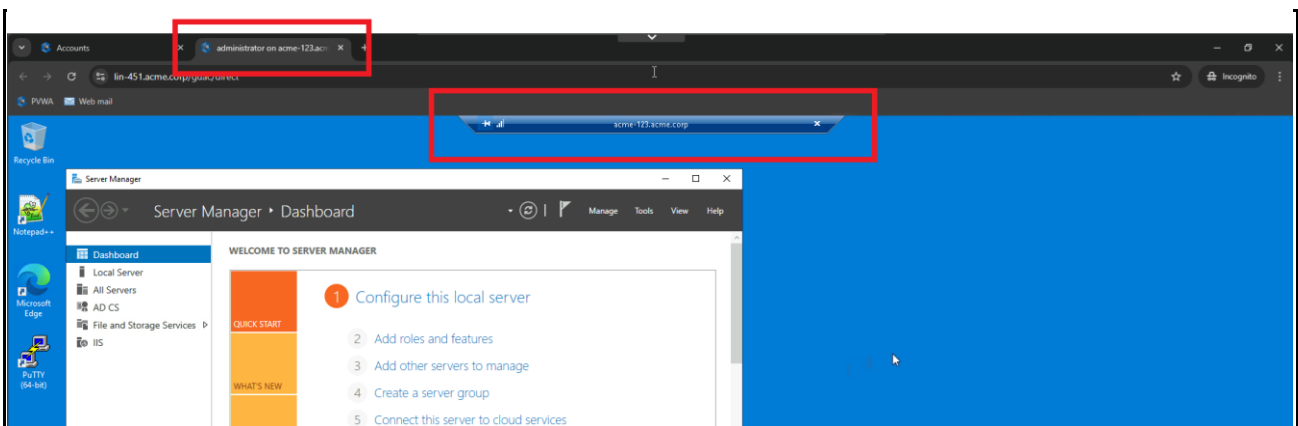
The image shows a 'Connect' dialog box with a close button (X) in the top right corner. It contains a 'Reason (Optional)' text box with the placeholder text 'Fix an issue'. Below this is a section titled 'Remote Connection Details' which is highlighted with a red rectangle. Inside this section, there is a toggle switch labeled 'Connect via browser' which is currently turned on. At the bottom right of the dialog is a blue 'Connect' button.

3. Press **Yes** to accept the host's RSA key, if asked.

Note: The ability to toggle between RDP file and HTML5GW connections is defined at the **Connection Component** level. For your convenience, the functionality has been preconfigured for RDP and SSH connections in this lab.

To enable this functionality for connection types other than RDP or SSH, go to **Options | Connection Components | PSM-RDP | User Parameters** and copy the **AllowSelectHTML5** parameter. Then paste it into a different connection component, for example: **PSM-WinSCP**

4. A new tab opens in the browser and you can see the RDP toolbar at the top. Keep the HTML5 connection open.



5. You can close this session either by closing the tab or by signing out. Make sure that you don't accidentally power off the server.

Transfer files via HTML5 GW – Optional exercise

In this section we will see how to copy a file from our workstation to the remote machine via the [HTML5 Gateway](#).

Enable drive mapping

For this exercise to work, you must first enable the drive mapping feature.

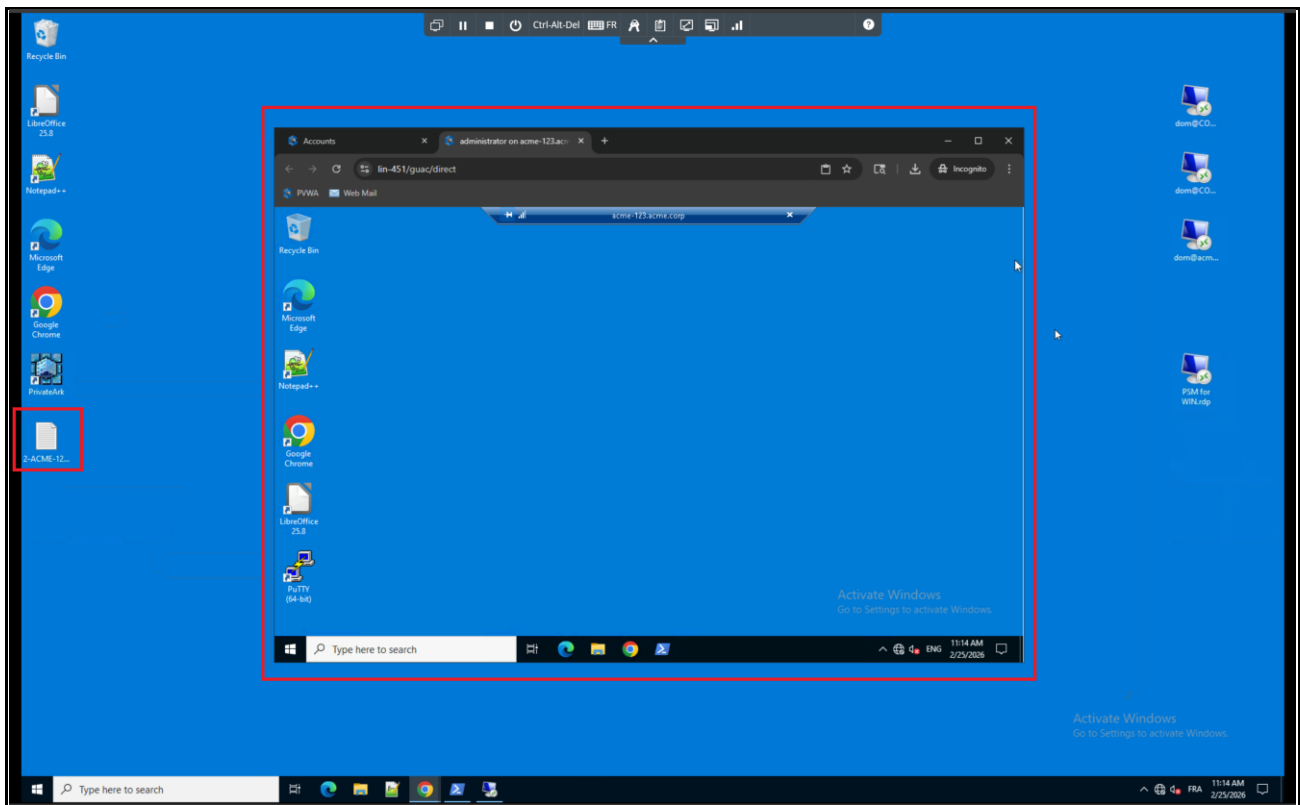
1. Log into the [PVWA](#) as *Mike* and navigate to: **Administration | Configuration Options | Options | Connection Components | PSM-RDP | User Parameters**. Select the parameter **AllowMappingLocalDrives**.
2. Enter the following values:

DisplayName	<i>Map local drives?</i>
Value	Yes
Visible	Yes

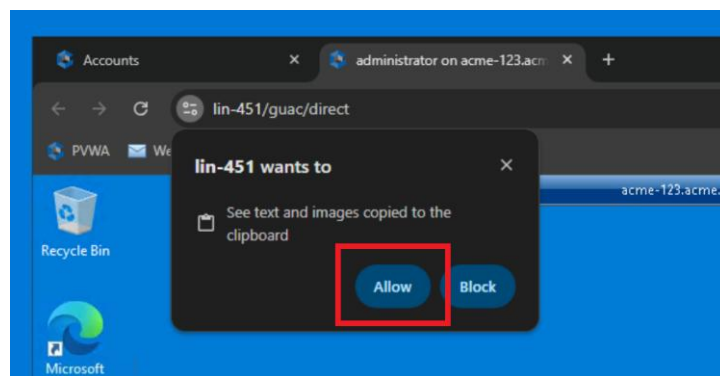
When you are finished, click **OK** to save the changes.

Test drive mapping

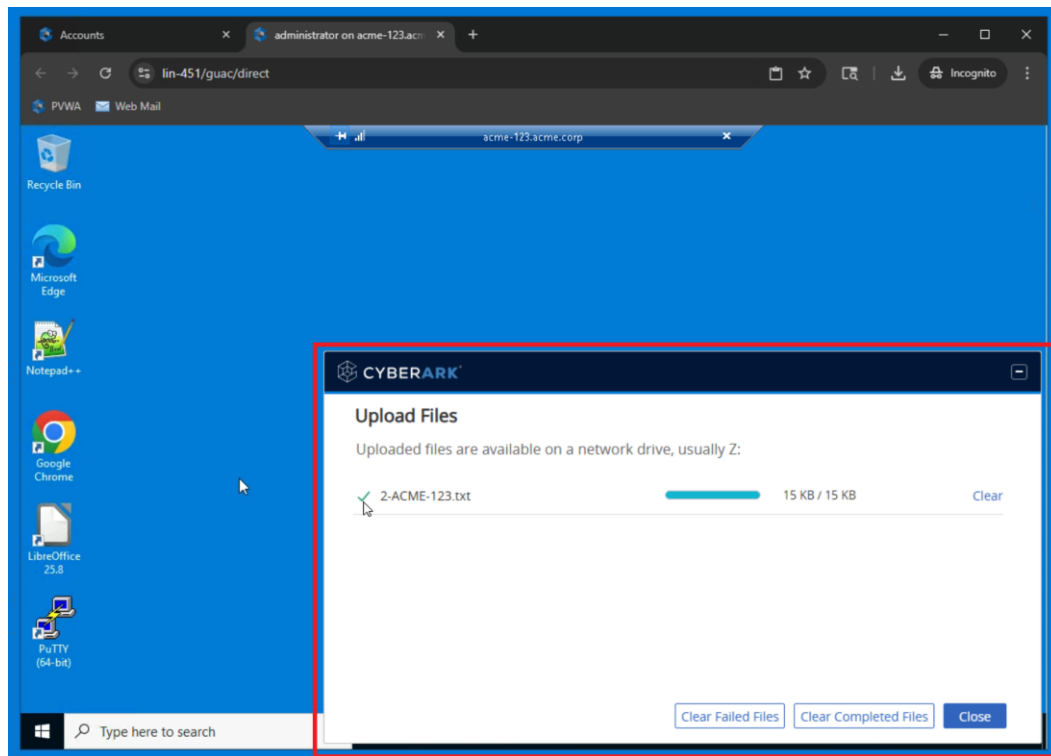
1. Relaunch *John's* connection with the *administrator* account and activate **Map local drives?**
2. Once connected, resize the Chrome browser window with the HTML5 connection in such a way that you can view the HTML5 session and the desktop for **COMP01**, as shown in the image below.



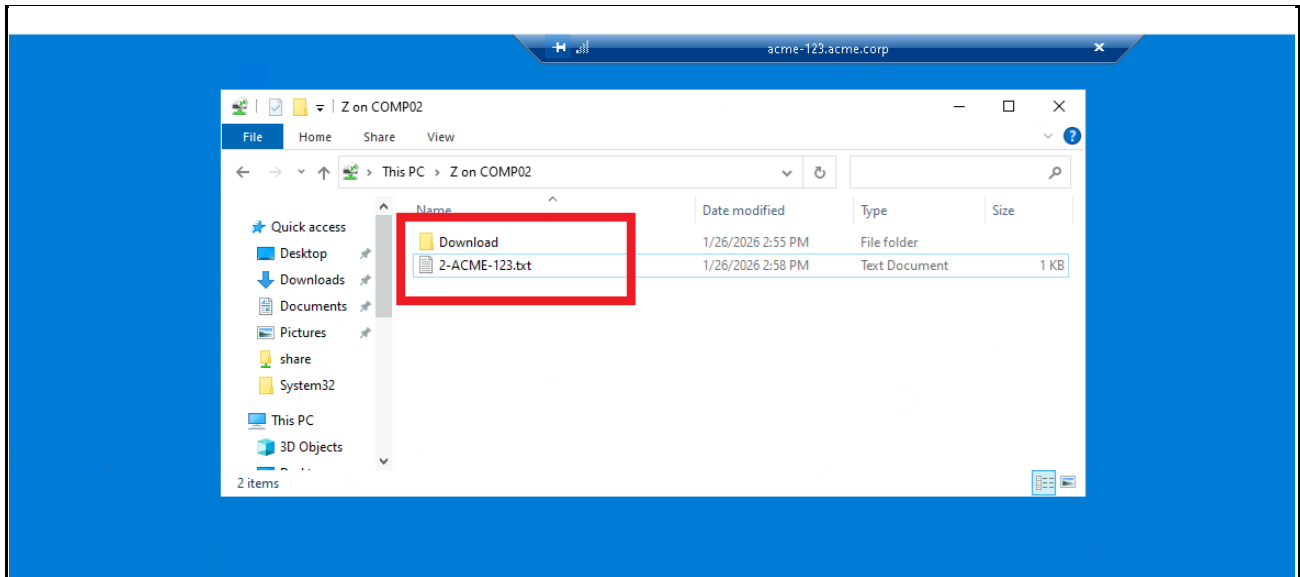
3. You will see a pop-up in Chrome asking you to allow the gateway server access to the clipboard. Click **Allow**.



4. On the **COMP01** desktop, you will find a file named **2-ACME-123.txt**. Drag and drop this file into the browser RDP window.
5. You should be able to see the following message stating that the file has been copied to the mapped drive **Z**, which you can view on the remote machine **ACME-123**. When the file transfer is finished, click on **Close**.

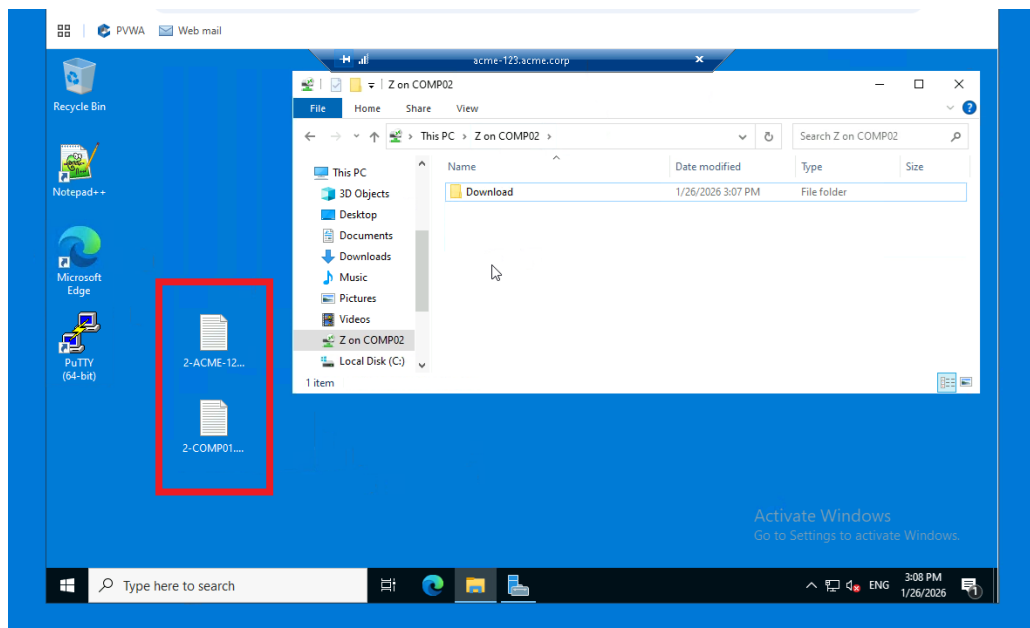


6. Next, inside the HTML5 session, open the file explorer, browse to the **z:** drive, and copy the file to the desktop on **ACME-123**.



Now we are going to copy a file in the other direction, from the remote machine back to our workstation.

7. Still working in the browser RDP window (so on **ACME-123**), make a copy of the file named **2-ACME-123.txt** that is now on the Desktop of **ACME-123**, and name it **2-COMP01.txt**.



8. Next, open the *Download* directory **Z on COMP01**. Drag and drop the **2-COMP01.txt** file that is on the desktop of the **ACME-123** into the *Download* directory.

Note: It will look like nothing has happened, but if you dropped the file to be transferred into the mapped *Download* directory, it has been transferred to the *Downloads* directory on the machine from which you initiated the connection. That is the **COMP01** server.

9. When you are finished, disconnect from the target server.

Connect using PSM Ad-Hoc Connection

Next, you will configure a **PSM Ad-Hoc Connection** (previously known as **Secure Connect**), which allows you to launch a **PSM** connection using unmanaged accounts. We will:

- Duplicate the **Platform PSM Secure Connect**, giving it an appropriate name.
- Activate the new **Platform**
- Activate the **PSM** for this **Platform** using an exception
- Test the Ad Hoc Connection

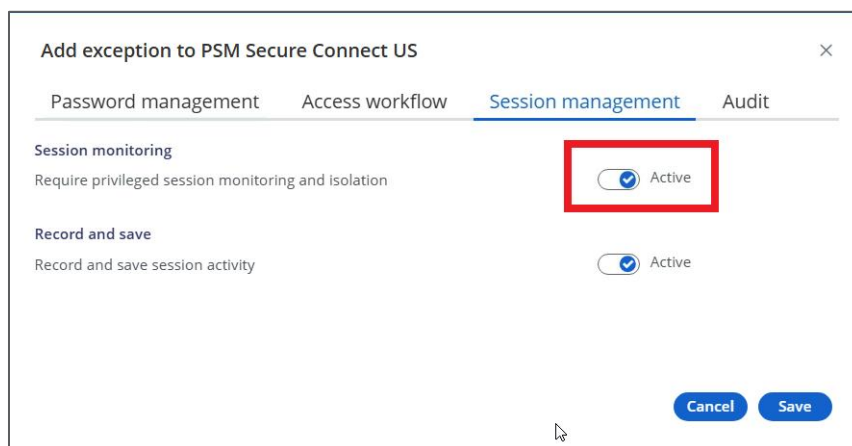
1. First, log into the **PVWA** as *Mike*, and go to **ADMINISTRATION | Platform Management**.
2. Locate the **Platform PSM Secure Connect** and duplicate it. Give it the name **PSM Secure Connect US**.

Hint: *PSM Secure Connect* is at the bottom of the list.

3. Because we have duplicated an inactive Platform, we will need to **activate** the new **Platform**.

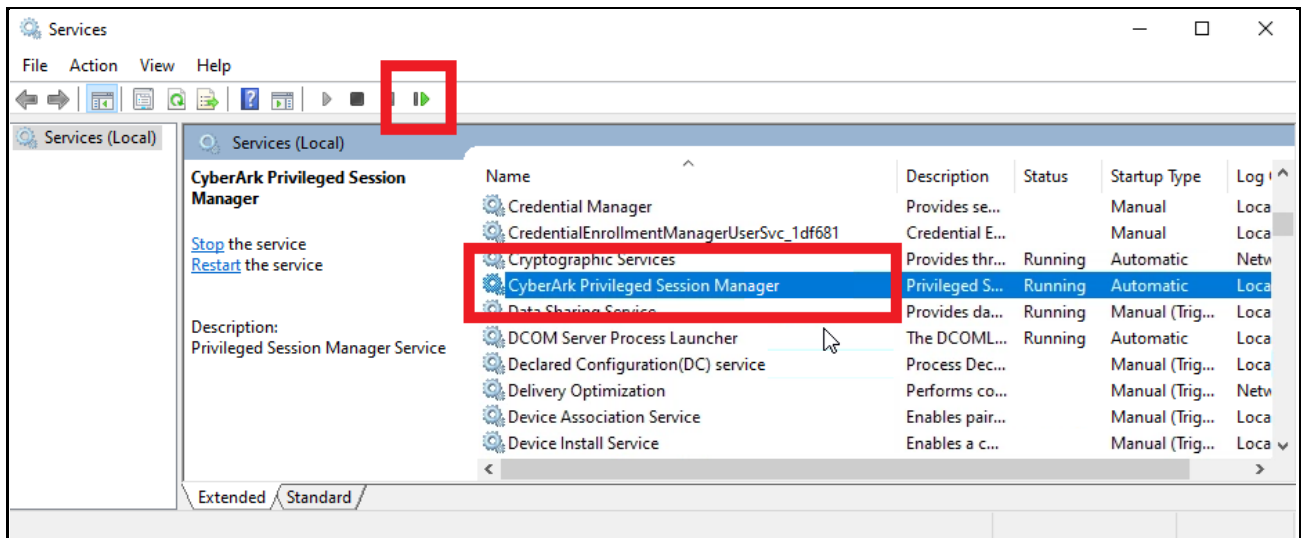
Note: As a matter of best practice, you should configure a dedicated recording Safe for this Platform.

4. Add an exception to the Master Policy by clicking on the ellipsis at the end of the line for the new Platform and clicking **Add exception**.
5. Click on the **Session management** tab and set the toggle for **Require privileged session monitoring and isolation** to **Active**. When you are ready, click **Save**.

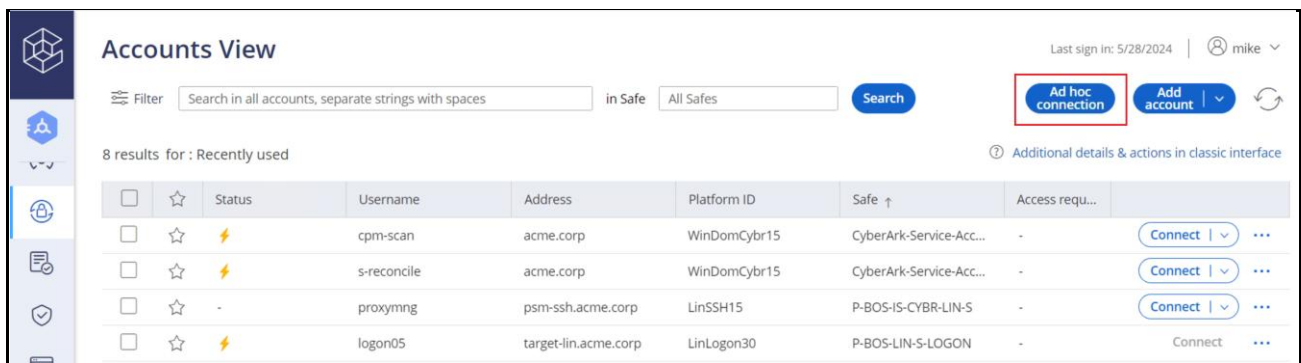


We could wait for the **PSM** to refresh its policies, but that could take up to 20 minutes. We can move things along by restarting the service.

6. To restart the **PSM** Service, you will need to connect to the server **COMP02**. There is an RDP shortcut on *Mike's* desktop for the user *dom* to connect to that machine. Once connected to **COMP02**, open the Windows Services app and restart the **CyberArk Privileged Session Manager** service.



7. Now go to the **ACCOUNTS** page and click on **Ad hoc connection**.



8. Enter the following and press **Connect**:

Platform	<i>PSMSecureConnectUS</i>
Client:	<i>WinSCP</i>
Address:	<i>10.0.21.2</i>
User Name:	<i>linuxusr01</i>
Password:	<i>Cyberark1</i>
Map Local Drives:	<i>Checked</i> (scroll down)
Port	<i>22</i>

Ad-Hoc connection

Platform

PSMSecureConnectUS

Client

WinSCP

Address

psm-gw.acme.corp

Username

linuxusr01

Password

.....

☒ Map local drives

Port

22

Reason (Optional)

Cancel

Connect

9. Press **Yes** to accept the host's RSA key.

Warning



Continue connecting to an unknown server and add its host key to a cache?

The server's host key was not found in the cache. You have no guarantee that the server is the computer you think it is.

The server's RSA key details are:

Algorithm: ssh-rsa 2048
 SHA-256: 3yjfUp2fAzDpWEkTR5slqWAlID0VQLfK/rRby7n55rc=
 MD5: a5:ed:aca6:b1:ff:38:27:78:08:da:0e:30:83:b2:55

If you trust this host, press Yes. To connect without adding host key to the cache, press No. To abandon the connection press Cancel.

[Copy key fingerprints to clipboard](#)

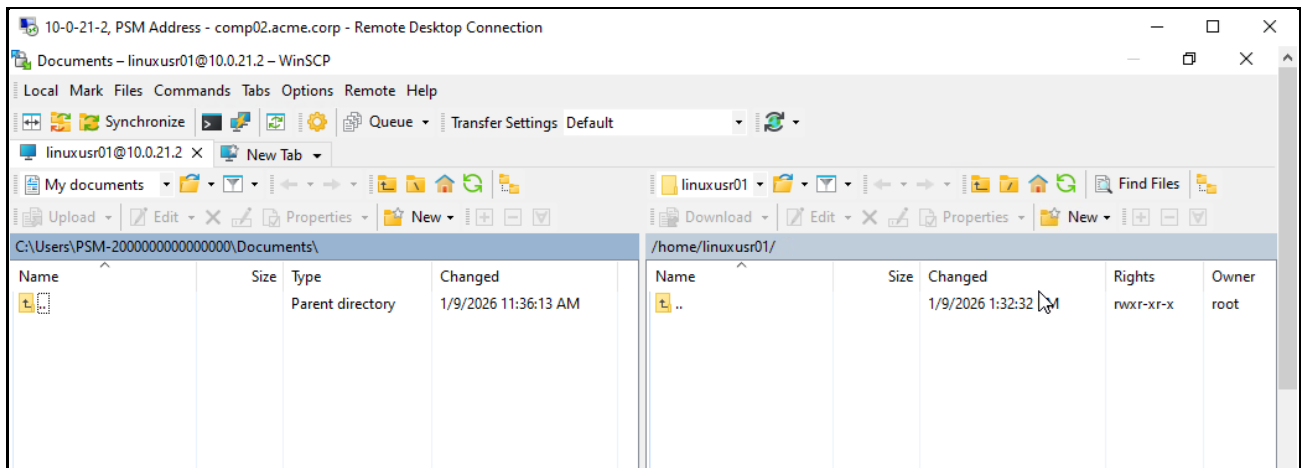
Yes

No

Cancel

Help

Optional: When you have connected to WinSCP, copy a file from the **PSM** server to the target machine. Suggestion: `C:\Add-Accounts\root-accounts.csv`



10. Press **F10** to exit and quit the application.

Privileged Session Manager for Windows

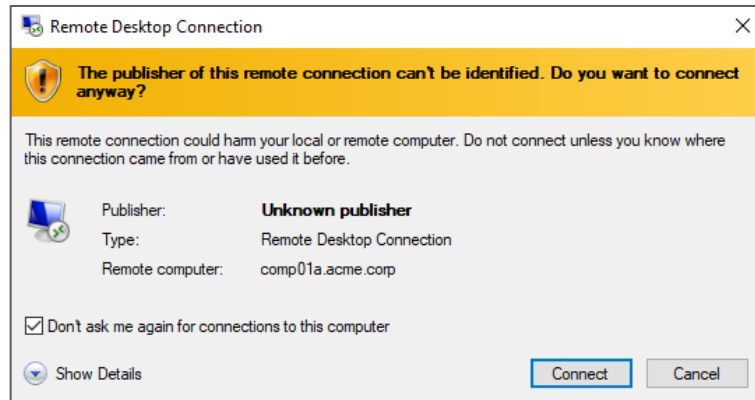
PSM for Windows (previously known as “PSM RDP Proxy”) enables users to connect through **PSM** to any remote target securely with a standard remote desktop client application like *mstsc* or an RDP connection manager.

You can also use preconfigured RDP files. When using RDP files, you can configure a single RDP file to connect through the **PSM** without providing the target system details or configure separate RDP files that include the target system details in advance. In this exercise we will look at both options for using preconfigured RDP files.

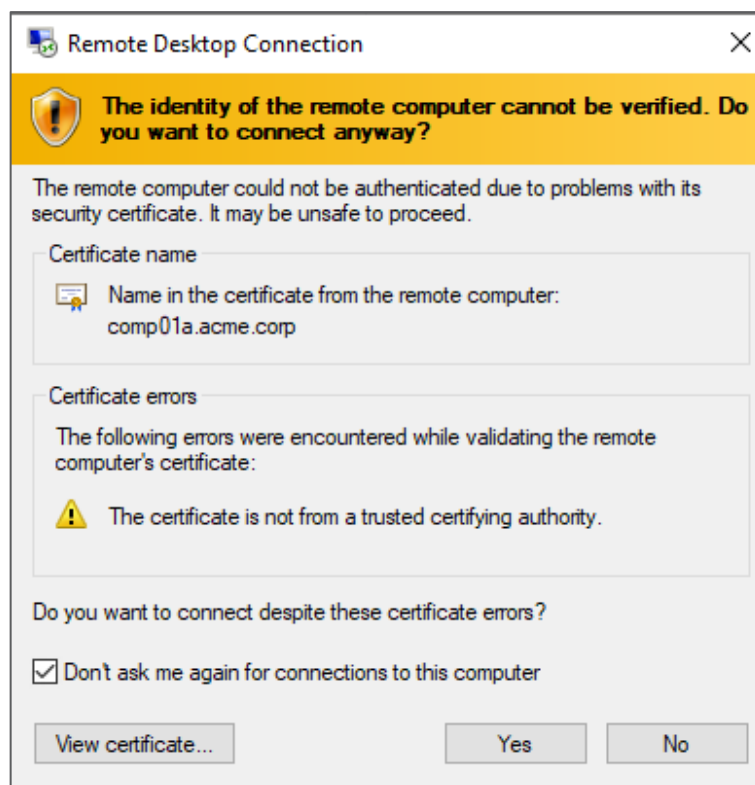
Connect using RDP file

In the first example, we will use a preconfigured RDP file without providing the target system details in advance. In this way, we are able to bypass the **PVWA**, but not **CyberArk** security.

1. On the desktop of the **comp01** server, you will find an RDP file named *PSM for WIN*. Double click on the file. When prompted, click on **Connect**.
2. Accept to run the application.

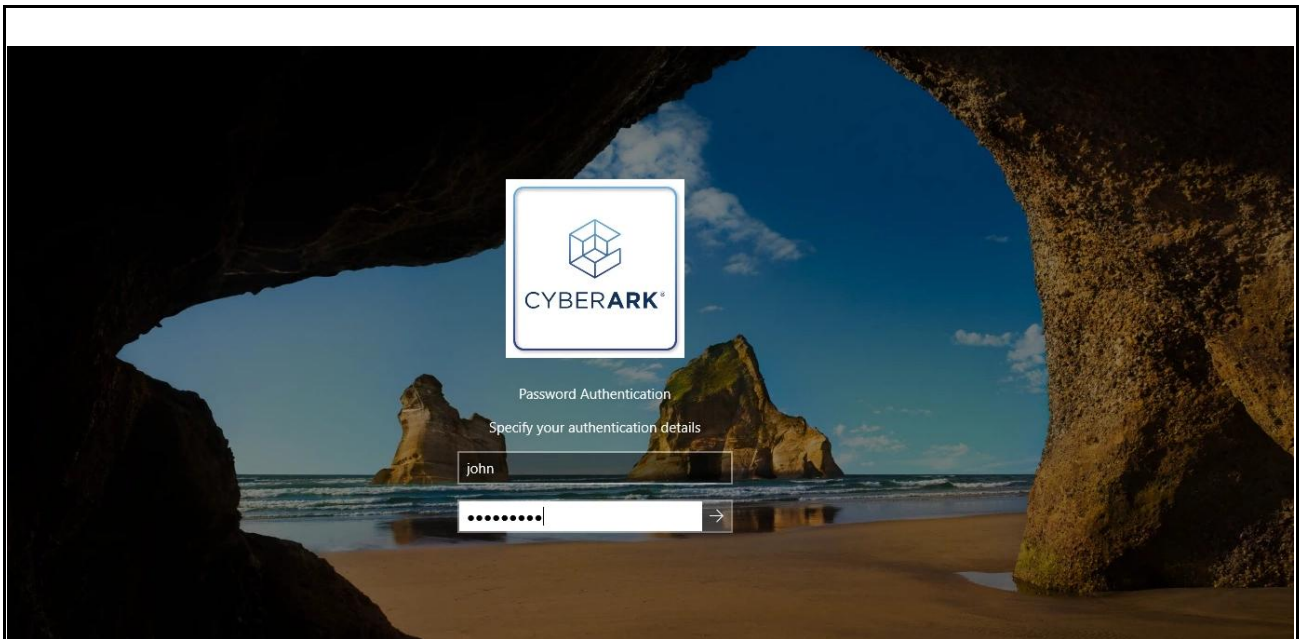


3. You may receive a certificate warning. Just click **Yes** to accept the certificate.



4. Enter the *John's* domain credentials:

Vault username: *John*
Password: *Cyberark1*



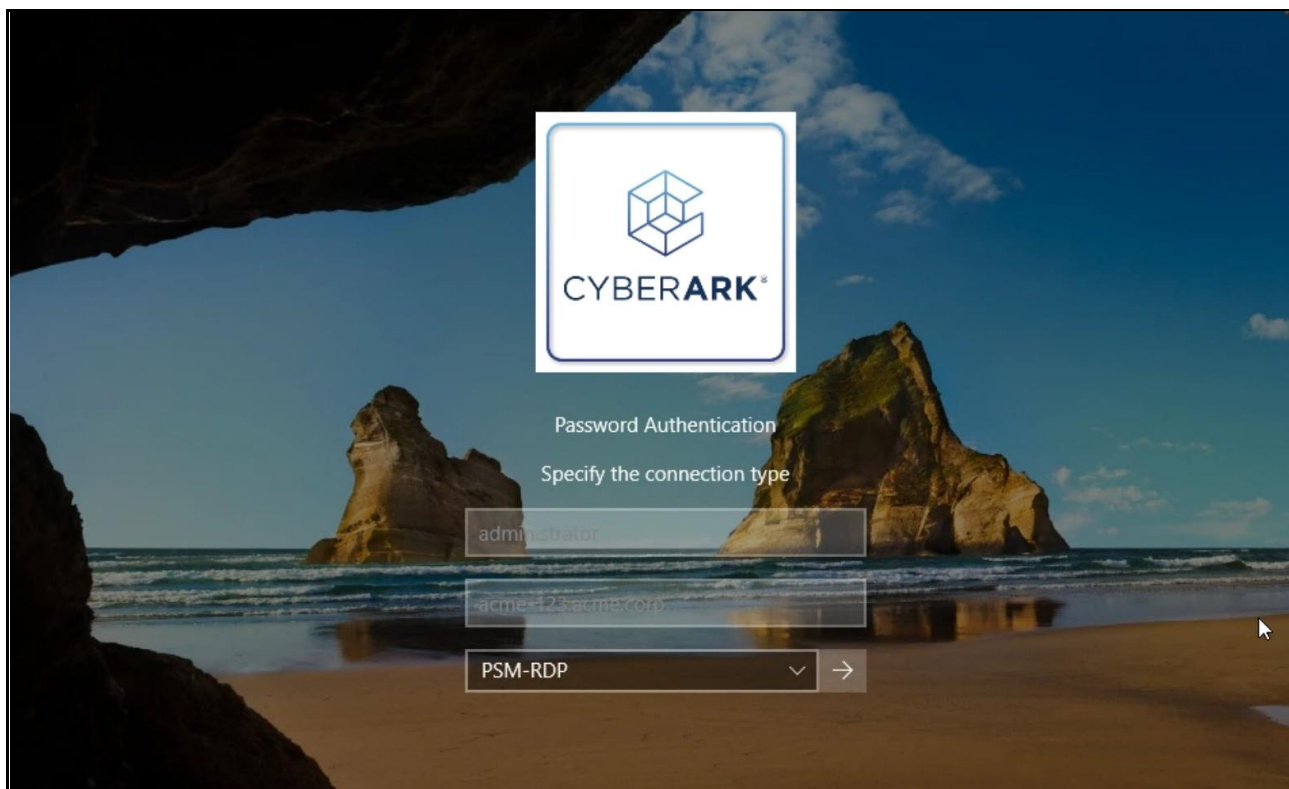
5. Next, input the target system details:

User Name:	<i>administrator</i>
Address:	<i>acme-123.acme.corp</i>

Note: The account name and address information must be provided here exactly as it is shown in **CyberArk PAM**. Because this account was created with the FQDN as the address, you cannot connect with the IP address or the short domain name.



6. Lastly, specify **PSM-RDP** as the connection type:



7. Confirm you were able to connect to the target system as *administrator*. Then disconnect (or sign out) from the target system.

Connect using RDP file with the target system details

We can make our jobs even easier by including the target details directly in the RDP file.

1. Right-click to open the **PSM for WIN** RDP file for editing using Notepad++.
2. Scroll to the bottom of the file. Note the two different alternate shells in the file. One is commented out, the other is active.
3. Edit the RDP file as follows to switch the active shells. This will include the target system details in advance. We can also insert a line providing *John's* user name so that on connection, *John* will only have to provide his LDAP password to connect. The lines should appear as below:

```
alternate shell:s:psm /u administrator /a acme-123.acme.corp /c PSM-RDP
username:s:john
# alternate shell:s:psm
```



```
40 shell working directory:s:  
41 gatewayhostname:s:  
42 gatewayusagemethod:i:4  
43 gatewaycredentialsource:i:4  
44 gatewayprofileusagemethod:i:0  
45 promptcredentialonce:i:0  
46 gatewaybrokeringtype:i:0  
47 use redirection server name:i:0  
48 rdgiskdcproxy:i:0  
49 kdcproxyname:s:  
50 alternate shell:s:psm /u administrator /a acme-123.acme.corp /c PSM-RDP  
51 username:s:john  
52 # alternate shell:s:psm  
53
```

4. Save the file and exit Notepad++.
5. Double-click on the RDP file to launch the connection. If configured properly, you will be prompted only for the Vault user's LDAP password. After you authenticate as *John*, the connection to the target machine as *administrator* will be made automatically.

Note: You can use any RDP client application to connect to any target system via the **PSM**. When setting up your RDP client, make sure to input the following details:

- *PSM Address*
- *Vault username*
- *RDP Start Program setting*

For more details on configuring RDP clients, please review the online [documentation](#).

Privileged Session Manager for SSH

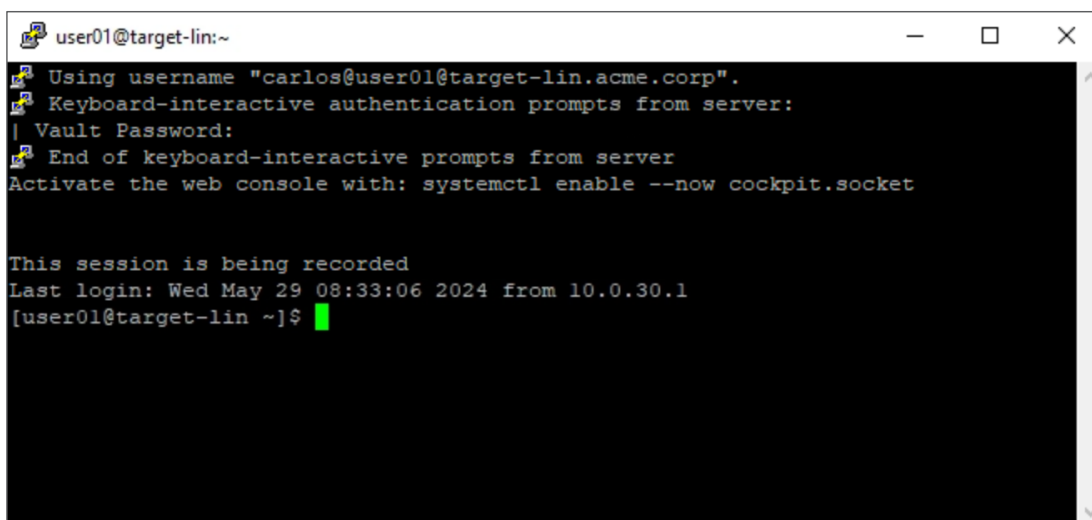
PSM for SSH (previously known as **PSM SSH Proxy** or **PSMP**) is designed to provide a native Unix/Linux user experience, connecting to any SSH target.

1. Open up a Powershell window on the **comp01** server. There is a shortcut in the Task bar.
2. Use the following connection string to connect to the **lin-451** machine using the **root01** account where the **Vault** user is *Carlos*.

```
ssh carlos@root01@lin-451.acme.corp@10.0.21.2
```

Hint: To be able troubleshoot any connection issues easily, make sure you mark “Never” under “Close window on exit”.

3. Enter **yes** to accept the server’s host key when prompted.
4. When prompted for a password, enter the password for *Carlos* (password: *Cyberark1*).



```

user01@target-lin:~
Using username "carlos@user01@target-lin.acme.corp".
Keyboard-interactive authentication prompts from server:
| Vault Password:
End of keyboard-interactive prompts from server
Activate the web console with: systemctl enable --now cockpit.socket

This session is being recorded
Last login: Wed May 29 08:33:06 2024 from 10.0.30.1
[user01@target-lin ~]$
  
```

5. Execute a few simple such as **pwd** and **ls -al** to generate some session activity. When you are done, enter **exit** and hit **Enter** to close the session.